

Curriculum Vitae: Jalim Singh, Ph.D.

CONTACT INFORMATION	Affiliation: Centre Inria de l'Université de Lorraine 615 Rue du Jardin-Botanique, 54600 Villers-lès-Nancy, France. Email: singhjalohsu@gmail.com, jalim.singh@inria.fr Webpages: LinkedIn, GitHub, Google Scholar
PROFESSIONAL SUMMARY	AI/ML Biophysicist with <i>PhD in Computational Polymer Physics</i> and 6+ years of postdoctoral experience, including last 3+ years in AI-driven protein dynamics and predictive cell biology. Experienced in developing and benchmarking generative AI models for protein dynamics prediction and cell biology relevant to drug discovery using EGNNs, VAEs, CNNs, Random Forest, GATs, GNNs, and Transformers. Proven track record in building scalable ML pipelines, integrating multi-modal data, and collaborating with interdisciplinary teams of biologists, physicists, and data scientists to advance biomedical innovation. Passionate about revolutionizing drug discovery for diseases including cancer, schizophrenia, and various neurological disorders by deploying and developing generative AI models.
RESEARCH INTERESTS	• AI-driven protein dynamics (ensemble) prediction relevant to drug discovery for diseases including cancers and neurological disorders. • Mechanistic and predictive modeling for cell biology using generative AI models including VAEs, Random Forest, GANs, GATs. • Collaborative innovation in interdisciplinary scientific teams.
SOFTWARE SKILLS	• Programming: Python, C, C++, MATLAB, LaTeX, BASH Scripting. • AI/ML Frameworks: PyTorch, Scikit-learn, Transformers, Equivariance Graph Neural Networks (EGNNs), Variational Auto-encoders (VAEs), Graph Neural Networks (GNNs), Convolutional Neural Networks (CNNs), Diffusion Models. • Software tools: Jupyter Notebook, Numpy, Scipy, Pandas, UMAP, Principal Component Analysis (PCA), Matplotlib, Seaborn. • Data Science: Multi-modal data integration, Predictive Modeling, Statistical Physics, Calculus, Linear Algebra, HPC cluster management. • Technical skills: Image processing – Cellpose 2, Image J, ilastik; Git version control. • Simulation Skills: Molecular Dynamics, Monte Carlo, and Brownian Dynamics Simulations using LAMMPS, OpenMM, GROMACS, and in house codes (samples on my GitHub). • Protein AI Expertise: Protein ensemble prediction, enzyme property modeling, molecular embeddings. • Domains: Generative AI for predictive biology and protein dynamics, drug-target interactions, scalable ML workflows.
OTHER SKILLS	• HPC cluster management, Scientific collaborations, communications, and writing. • Member of the inner organizing committee in a conference on “Recent Topics in Statistical Mechanics”, held at National Institute of Science Education and Research Bhubaneswar, India in December 2019.
EDUCATION	Ph.D., School of Basic Sciences, Indian Institute of Technology Mandi, Himachal Pradesh, India, <i>Degree awarded:</i> October 2018 • Thesis Topic: <i>Microscopic structure and dynamics of glass transition under phase separation in a model linear polymer melt</i>, Web link

- Conducted Molecular Dynamics, Monte Carlo, and Brownian Dynamics simulations of polymers followed by analyzing results from concepts of Statistical Physics.
- Advisor: [Dr. Prasanth P. Jose, Ph.D.](#)
- CGPA (During course work): 9.0

M.Sc. in Physics, Department of Physics, Banaras Hindu University, Varanasi, Uttar Pradesh, India, June 2011

Key Courses: Advanced Quantum Mechanics, Advanced Statistical Mechanics, Mathematical Physics, Numerical Analysis & Programming; **CGPA** : 8.16

AWARDS/HONOURS • Best oral presentation at “Research Fair 2017”, Indian Institute of Technology Mandi, India.

RESEARCH EXPERIENCE **Advanced Researcher** January 2026 – present.
 Centre Inria, University of Lorraine, Nancy, France
Advisor: Dr. Yasaman Karami

Key Contributions:

- Developing deep learning model for allosteric pathways predictions.

Postdoctoral Fellow October 2024 – November 2025.
 Department of Pharmacology, Center for AI in Protein Dynamics
 Vanderbilt University, Nashville, TN, USA
Advisor: Dr. Benjamin P. Brown

Key Contributions:

- Fine tuned generative AI models (GNNs, GATs, EGNNs, VAEs, diffusion-inspired) for training using PyTorch to predict protein conformational dynamics.
- Performed a benchmark of recent GenAI models ([JAMUN](#), [TRAJCAST](#), [TITO](#)) for molecular dynamics predictions at the longer time scales.
- Designed scalable ML pipelines, leveraging atom-type specific embeddings, crucial for computer-aided drug design.

Postdoctoral Fellow October 2022 – September 2024.
 Department of Biomedical Engineering
 Oregon Health & Science University, Portland OR, USA.
Advisor: Daniel M. Zuckerman, Ph.D.

Key Contributions:

- Built predictive models by integrating multi-modal biological data (RNA-seq, live-cell imaging) for predicting RNA levels of genes and cell-cycle dynamics.
- Optimized high-throughput ML/AI workflows in Python on HPC cluster, enhancing model scalability and performance.
- Collaborated with data scientists and experimentalists to refine CNN and Transformer-based architectures for biological applications.

Postdoctoral Fellow January 2021 – April 2022
 Central European Institute of Technology (CEITEC)
 Masaryk University, Brno, Czech Republic
 Advisor: Dr. Robert Vácha, Ph.D. *Key Contributions:*

- Performed coarse-grained MD and Monte Carlo simulations of the CTD of RNA Pol II, showing phase separation, important in gene transcription mechanism.

Postdoctoral Fellow

December 2018 – December 2020

School of Physical Sciences

National Institute of Science Education and Research Bhubaneswar, Odisha, India

Advisor: A. V. Anil Kumar, Ph.D.

Assistant Professor

December 2017 – December 2018

Department of Physics, Sandip University, Nashik, Maharashtra, India

Courses taught: Classical Mechanics, Nuclear Physics, Statistical Mechanics, Programming lab.

SELECTED PUBLICATIONS

1. **J. Singh** and Benjamin P Brown, “What generative AI models can and can’t do to predict molecular dynamics?”, *under preparation*.
2. **J. Singh**, J. Copperman, L. M. Heiser, D. M. Zuckerman, “Inferring single-cell dynamics of a molecular reporter from unlabeled live-cell light microscopy data analyzed with delay embedding”, [bioRxiv](#) 2025.
3. D. C. Thakur, **J. Singh**, and A. V. A. Kumar, “Phase separation in a binary mixture of sticky spheres”, *J. Chem. Phys.* **163**, 024904 (2025).
4. J. Copperman, I. C. Mclean, S. M. Gross, **J. Singh**, Y. Hwan Chang, D. M. Zuckerman, L. M. Heiser, “Single-cell morphodynamical trajectories enable prediction of gene expression accompanying cell state change”, *Cell Systems* **17**, 101567 (2026).
5. **J. Singh**, M. Mustakim and A. V. A. Kumar “Super-Arrhenius diffusion in a binary colloidal mixture at low volume fraction: an effect of depletion interaction due to an asymmetric barrier”, *J. Phys.: Condens. Matter* **33**, 125101 (2021).
6. **J. Singh** and P. P. Jose “Violation of Stokes-Einstein and Stokes-Einstein-Debye relations in polymers at the gas-supercooled liquid coexistence”, *J. Phys.: Condens. Matter* **33**, 055401 (2021).
7. **J. Singh** and A. V. A. Kumar “Phase separation in a two-dimensional binary colloidal mixture by quorum sensing activity”, *Phys. Rev. E* **101**, 022606 (2020).
8. **J. Singh** and P. P. Jose “Heterogeneous dynamics in linear polymer melts near glass transition temperature”. *J. Phys.: Conf. Ser.* **759**, 012018 (2016).

SELECTED PRESENTATIONS

Conferences/Workshops

- Presented at “Summer RosettaCon 2025”, Seattle, Washington, USA August 2025.
- Presented at “Chemistry and Pharmacology of Drug Abuse”, Boston, Massachusetts, USA August 2025.
- Poster & Oral presentation at “Biophysical Society (BPS)” meeting, Philadelphia, Pennsylvania, USA February 2024.
- Oral presentation at “Recent Topics in Statistical Mechanics”, National Institute of Science Education and Research, Bhubaneswar, India December 2019
- Oral & poster presentation at “Department of Atomic Energy Solid State Physics Symposium-2019”, organized by Bhabha Atomic Research Center (BARC) Mumbai at Indian Institute of Technology, Jodhpur, India December 2019
- Poster presentation at “Recent Advances in Dynamics at the Interface of Chemistry and Biology”, Indian Institute of Science Bangalore, India February 2019

- Poster presentation at “Dynamics of glass forming liquids: will theory and experiment ever meet?”, Glass and Time Research, Roskilde University, Roskilde, Denmark April 2017
- Poster & Oral presentation at “Complex Fluids2016”, Indian Institute of Information Technology, Hyderabad, India December 2016

REFERENCES

Benjamin P. Brown, MD, Ph.D.

Assistant Professor

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Vanderbilt University, Nashville TN, 37212, USA.

Daniel M. Zuckerman, Ph.D.

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Biomedical Engineering, School of Medicine

E-mail: zuckerm@ohsu.edu

Oregon Health & Science University, Portland OR, USA.

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School of Basic Sciences

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Indian Institute of Technology Mandi, India

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